

Error and Decomposition Metrics

Metric	Definition	Form
Residual Sum of Squares (RSS)	Measures the unexplained variation in the dependent variable after fitting the regression model.	RSS
Total Sum of Squares (TSS)	Measures the total variation in the dependent variable around its mean.	TSS
Model Sum of Squares (MSS) or Regression Sum of Squares (ESS)	Measures the variation explained by the regression model.	MSS
Mean Absolute Error (MAE)	Average of absolute differences between actual and predicted values.	MAE

Relationship

$$TSS = MSS + RSS$$